Chapter 3

Algebra Primer

The basic geometric objects that are studied in algebraic geometry are algebraic varieties. An algebraic variety is just the set of solutions to a system of polynomial equations. On the algebraic side, varieties will form one of the main objects of study in this book. In particular, we will be interested in ways to use algebraic and computational techniques to understand geometric properties of varieties. This chapter will be devoted to laying the groundwork for algebraic geometry. We introduce algebraic varieties and the tools used to study them, in particular, their vanishing ideals and Gröbner bases of these ideals. In Section 3.4 we will describe some applications of Gröbner bases to computing features of algebraic varieties. Our view is also that being able to compute key quantities in the area is important, and we illustrate the majority of concepts with computational examples using computer algebra software.

Our aim in this chapter is to provide a concise introduction to the main tools we will use from computational algebra and algebraic geometry. Thus, it will be highly beneficial for the reader to also consult some texts that go into more depth on these topics, for instance [**CLO07**] or [**Has07**].

3.1. Varieties

Let $\mathbb{K}$ be a field. For the purposes of this book, there are three main fields of interest: the rational numbers $\mathbb{Q}$, which are useful for performing computations, the real numbers $\mathbb{R}$, which are fundamental because probabilities are real, and the complex numbers $\mathbb{C}$, which we will need in some circumstances for proving precise theorems in algebraic geometry.

Let $p_1, p_2, \ldots, p_r$ be polynomial variables or *indeterminates* (we will generally use the word indeterminate instead of variable to avoid confusion with random variables). A *monomial* in these indeterminates is an expression of the form

$$p^u := p_1^{u_1} p_2^{u_2} \cdots p_r^{u_r}, \tag{3.1.1}$$

where $u = (u_1, u_2, \ldots, u_r) \in \mathbb{N}^r$ is a vector of nonnegative integers. The set of integers is denoted by $\mathbb{Z}$ and the set of nonnegative integers is $\mathbb{N}$. The notation p^u is monomial vector notation for the monomial in equation (3.1.1).

A *polynomial* in indeterminates $p_1, p_2, \ldots, p_r$ over $\mathbb{K}$ is a linear combination of finitely many monomials, with coefficients in $\mathbb{K}$. That is, a polynomial can be represented as a sum

$$f = \sum_{u \in A} c_u p^u,$$

where A is a finite subset of $\mathbb{N}^r$ and each coefficient $c_u \in \mathbb{K}$. Alternatively, we could say that $f = \sum_{u \in \mathbb{N}^r} c_u p^u$, where only finitely many of the coefficients c_u are nonzero elements of the field $\mathbb{K}$.

For example, the expression $p_1^2 p_3^4$ is a monomial in the indeterminates p_1, p_2, p_3, with $u = (2, 0, 4)$, whereas $p_1^2 p_3^{-4}$ is not a monomial because there is a negative in the exponent of indeterminate p_3. An example of a polynomial is $f = p_1^2 p_3^4 + 6p_1^2 p_2 - \sqrt{2} p_1^3 p_2 p_3^2$ which has three terms. The power series

$$\sum_{i=0}^{\infty} p_1^i$$

is not a polynomial because it has infinitely many terms with nonzero coefficients.

Let

$$\mathbb{K}[p] := \mathbb{K}[p_1, p_2, \ldots, p_r]$$

be the set of all polynomials in the indeterminates $p_1, p_2, \ldots, p_r$ with coefficients in $\mathbb{K}$. This set of polynomials forms a *ring*, because the sum of two polynomials is a polynomial and the product of two polynomials is a polynomial; these two operations follow all the usual associative, distributive, and commutative properties of addition and multiplication. Each polynomial is naturally considered as a multivariate function from $\mathbb{K}^r \to \mathbb{K}$, by evaluation. For this reason, we will often refer to the ring $\mathbb{K}[p]$ as the *ring of polynomial functions*.

Definition 3.1.1. Let $S \subseteq \mathbb{K}[p]$ be a set of polynomials. The *variety* defined by S is the set

$$V(S) := \{a \in \mathbb{K}^r : f(a) = 0 \text{ for all } f \in S\}.$$

The variety $V(S)$ is also called the *zero set* of S.

Note that the set of points in the variety depends on which field $\mathbb{K}$ we are considering. Most of the time, the particular field of interest should be clear from the context. In nearly all situations explored in this book, this will either be the real numbers $\mathbb{R}$ or the complex numbers $\mathbb{C}$. When we wish to call attention to the nature of the particular field under consideration, we will use the notation $V_{\mathbb{K}}(S)$ to indicate the variety in $\mathbb{K}^r$.

Example 3.1.2 (Some varieties over $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$)**.** The variety $V(\{p_1 - p_2^2\}) \subset \mathbb{R}^2$ is the familiar parabolic curve, facing towards the right, whereas the variety $V(\{p_1^2+p_2^2-1\}) \subset \mathbb{R}^2$ is a circle centered at the origin. The variety $V(\{p_1-p_2^2, p_1^2+p_2^2-1\})$ is the set of points that satisfy both of the equations

$$p_1 - p_2^2 = 0, \quad p_1^2 + p_2^2 - 1 = 0$$

and thus is the intersection $V(\{p_1-p_2^2\}) \cap V(\{p_1^2+p_2^2-1\})$. Thus, the variety $V(\{p_1 - p_2^2, p_1^2 + p_2^2 - 1\})$ is a finite number of points. It makes a difference which field we consider this variety over, and the number of points varies with the field. Indeed,

$$V_{\mathbb{Q}}(\{p_1 - p_2^2, p_1^2 + p_2^2 - 1\}) = \emptyset,$$

$$V_{\mathbb{R}}(\{p_1 - p_2^2, p_1^2 + p_2^2 - 1\}) = \left\{ \left(\frac{-1+\sqrt{5}}{2}, \pm\sqrt{\frac{-1+\sqrt{5}}{2}} \right) \right\},$$

$$V_{\mathbb{C}}(\{p_1 - p_2^2, p_1^2 + p_2^2 - 1\}) = \left\{ \left(\frac{-1\pm\sqrt{5}}{2}, \pm\sqrt{\frac{-1\pm\sqrt{5}}{2}} \right) \right\}.$$

In many situations, it is natural to consider polynomial equations like $p_1^2 + 4p_2 = -17$, that is, polynomial equations that are not of the form $f(p) = 0$, but instead of the form $f(p) = g(p)$. These can, of course, be transformed into the form discussed previously, by moving all terms to one side of the equation, i.e., $f(p) - g(p) = 0$. Writing equations in this standard form is often essential for exploiting the connection to algebra which is at the heart of algebraic geometry.

In many situations in algebraic statistics, varieties are presented as parametric sets. That is, there is some polynomial map $\phi : \mathbb{K}^d \to \mathbb{K}^r$ such that $V = \{\phi(t) : t \in \mathbb{K}^d\}$ is the image of the map. We will also use $\phi(\mathbb{K}^d)$ as shorthand to denote this image.

Example 3.1.3 (Twisted cubic)**.** Consider the polynomial map

$$\phi : \mathbb{C} \to \mathbb{C}^3, \qquad \phi(t) = \left(t, t^2, t^3\right).$$

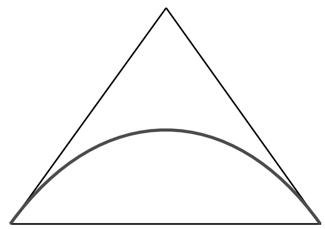

Figure 3.1.1. The set of all binomial distributions for $r = 2$ trials.

The image of this map is the *twisted cubic curve* in $\mathbb{C}^3$. Every point on the twisted cubic satisfies the equations $p_1^2 = p_2$ and $p_1^3 = p_3$, and conversely every point satisfying these two equations is in the image of the map ϕ.

The next example concerns a parametric variety which arises naturally in the study of statistical models.

Example 3.1.4 (Binomial random variables)**.** Consider the polynomial map $\phi : \mathbb{C} \to \mathbb{C}^{r+1}$ whose ith coordinate function is

$$\phi_i(t) = \binom{r}{i} t^i (1-t)^{r-i}, \qquad i = 0, 1, \ldots, r.$$

For a real parameter $\theta \in [0, 1]$, the value $\phi_i(\theta)$ is the probability $P(X = i)$, where X is a binomial random variable with success probability θ. That is, if θ is the probability of getting a head in one flip of a biased coin, $\phi_i(\theta)$ is the probability of seeing i heads in r independent flips of the same biased coin. Hence, the vector $\phi(\theta)$ denotes the probability distribution of a binomial random variable.

The image of the real interval, $[0, 1]$, is a curve inside the *probability simplex*

$$\Delta_r = \left\{ p \in \mathbb{R}^{r+1} \ : \ \sum_{i=0}^{r} p_i = 1, p_i \geq 0 \text{ for all } i \right\}$$

which consists of all possible probability distributions of a binomial random variable. For example, in the case of $r = 2$, the curve $\phi([0, 1])$ is the curve inside the probability simplex in Figure 3.1.1.

The image of the complex parametrization $\phi(\mathbb{C})$ is a curve in $\mathbb{C}^{r+1}$ that is also an algebraic variety. In particular, it is the set of points satisfying the trivial equation $\sum_{i=0}^{r} p_i = 1$ together with the vanishing of all 2×2 subdeterminants of the following $2 \times r$ matrix:

$$\begin{pmatrix} p_0/\binom{r}{0} & p_1/\binom{r}{1} & p_2/\binom{r}{2} & \cdots & p_{r-1}/\binom{r}{r-1} \\ p_1/\binom{r}{1} & p_2/\binom{r}{2} & p_3/\binom{r}{3} & \cdots & p_r/\binom{r}{r} \end{pmatrix}. \tag{3.1.2}$$

In applications to algebraic statistics, it is common to consider not only maps whose coordinate functions are polynomial functions, but also maps whose coordinate functions are rational functions. Such functions have the form f/g, where f and g are polynomials. Hence, a *rational map* is a map $\phi : \mathbb{C}^d \dashrightarrow \mathbb{C}^r$, where each coordinate function $\phi_i = f_i/g_i$; here $f_i, g_i \in \mathbb{K}[t] := \mathbb{K}[t_1, \ldots, t_d]$ are polynomials. Note that the map ϕ is not defined on all of $\mathbb{C}^d$, it is only well-defined off of the hypersurface $V(\{\prod_i g_i\})$. If we speak of the image of ϕ, we mean the image of the set $\mathbb{C}^d \setminus V(\{\prod_i g_i\})$, where the rational map ϕ is well-defined. The symbol $\dashrightarrow$ is used in the notational definition of a rational map to indicate that the map need not be defined everywhere.

3.2. Ideals

In the preceding section, we showed how to take a collection of polynomials (an algebraic object) to produce their common zero set (a geometric object). This construction can be reversed: from a geometric object (subset of $\mathbb{K}^r$) we can construct the set of polynomials that vanish on it. The sets of polynomials that can arise in this way have the algebraic structure of a radical ideal. This interplay between varieties and ideals is at the heart of algebraic geometry. We describe the ingredients in this section.

Definition 3.2.1. Let $W \subseteq \mathbb{K}^r$. The set of polynomials

$$I(W) \quad = \quad \{f \in \mathbb{K}[p] : f(a) = 0 \text{ for all } a \in W\}$$

is the *vanishing ideal* of W or the *defining ideal* of W.

Recall that a subset I of a ring R is an *ideal* if $f + g \in I$ for all $f, g \in I$ and $hf \in I$ for all $f \in I$ and $h \in R$. As the name implies, the set $I(W)$ is an ideal (e.g., Exercise 3.1).

A collection of polynomials $\mathcal{F}$ is said to *generate* an ideal I if it is the case that every polynomial $f \in I$ can be written as a finite linear combination of the form $f = \sum_{i=1}^k h_i f_i$, where the $f_i \in \mathcal{F}$ and the h_i are arbitrary polynomials in $\mathbb{K}[p]$. number k might depend on f. In this case we use the notation $\langle \mathcal{F} \rangle = I$ to denote that I is generated by $\mathcal{F}$.

Theorem 3.2.2 (Hilbert basis theorem)**.** *For every ideal I in a polynomial ring $\mathbb{K}[p_1, \ldots, p_r]$ there exists a finite set of polynomials $\mathcal{F} \subset I$ such that $I = \langle \mathcal{F} \rangle$. That is, every ideal in $\mathbb{K}[p]$ has a finite generating set.*

Note that the Hilbert basis theorem only holds for polynomial rings in finitely many variables. In general, a ring that satisfies the property that every ideal has a finite generating set is called a Noetherian ring.

When presented with a set $W \subseteq \mathbb{K}^r$, we are often given the task of finding a finite set of polynomials in $I(W)$ that generate the ideal. The special case where the set $W = \phi(\mathbb{K}^d)$ for some rational map ϕ is called an *implicitization problem* and many problems in algebraic statistics can be interpreted as implicitization problems.

Example 3.2.3 (Binomial random variables)**.** Let $\phi : \mathbb{C} \to \mathbb{C}^{r+1}$ be the map from Example 3.1.4. Then the ideal $I(\phi(\mathbb{C})) \subseteq \mathbb{C}[p_0, \ldots, p_r]$ is generated by the polynomials $\sum_{i=0}^{r} p_i - 1$ plus the 2×2 minors of the matrix from (3.1.2).

In the special case where $r = 2$, we have

$$I(\phi(\mathbb{C})) = \langle p_0 + p_1 + p_2 - 1, 4p_0p_2 - p_1^2 \rangle.$$

So the two polynomials $p_0+p_1+p_2-1$ and $4p_0p_2-p_1^2$ solve the implicitization problem in this instance. We will describe computational techniques for solving the implicitization problem in Section 3.4.

If we are given an ideal, we can iteratively compute the variety defined by the vanishing of that ideal, and then compute the vanishing ideal of the resulting variety. We always have $I \subseteq I(V(I))$. However, this containment need not be an equality, as the example $I = \langle p_1^2 \rangle$ indicates. The ideals that can arise as vanishing ideals are forced to have some extra structure.

Definition 3.2.4. An ideal I is called *radical* if $f^k \in I$ for some polynomial f and positive integer k implies that $f \in I$. The radical of I, denoted $\sqrt{I}$, is the smallest radical ideal that contains I and consists of the polynomials:

$$\sqrt{I} \quad = \quad \left\{ f \in \mathbb{K}[p] : f^k \in I \text{ for some } k \in \mathbb{N} \right\}.$$

The following straightforward proposition explains the importance of radical ideals in algebraic geometry. Its proof is left to the exercises.

Proposition 3.2.5. *For any field and any set $W \subset \mathbb{K}^r$, the ideal $I(W)$ is a radical ideal.*

A much more significant result shows that the process of taking vanishing ideals of varieties gives the radical of an underlying ideal, in the case of an algebraically closed field. (Recall that a field is *algebraically closed* if every univariate polynomial with coefficients in that field has a root in the field.)

Theorem 3.2.6 (Nullstellensatz)**.** *Suppose that $\mathbb{K}$ is algebraically closed (e.g., $\mathbb{C}$). Then the vanishing ideal of the variety of an ideal is the radical of the ideal: $I(V(I)) = \sqrt{I}$.*

See [**CLO07**] or [**Has07**] for a proof of Theorem 3.2.6. When $\mathbb{K}$ is algebraically closed, the radical provides a closure operation for ideals, by determining all the functions that vanish on the underlying varieties $V(I)$.

Computing $V(I(W))$ provides the analogous algebraic closure operations for subsets of $\mathbb{K}^r$.

Proposition 3.2.7. *The set $V(I(W))$ is called the* Zariski closure *of W. It is the smallest algebraic variety that contains W.*

See Exercise 3.1 for the proof of Propositions 3.2.5 and 3.2.7. A set $W \subseteq \mathbb{K}^r$ is called *Zariski closed* if it is a variety that is of the form $W = V(S)$ for some $S \subseteq \mathbb{K}[p]$. Note that since polynomial functions are continuous in the standard Euclidean topology, the Zariski closed sets are also closed in the standard Euclidean topology.

Example 3.2.8 (A simple Zariski closure)**.** As an example of a Zariski closure let $W \subset \mathbb{R}^2$ consist of the p_1-axis with the origin $(0,0)$ removed. Because polynomials are continuous, any polynomial that vanishes on W vanishes on the origin as well. Thus, the Zariski closure contains the p_1-axis. On the other hand, the p_1-axis is the variety defined by the ideal $\langle p_2 \rangle$, so the Zariski closure of W is precisely the p_1-axis.

Note that if we consider the same set W but over a finite field $\mathbb{K}$, the set W will automatically be Zariski closed (since every finite set is always Zariski closed).

Example 3.2.9 (Binomial random variable)**.** Let $\phi : \mathbb{C} \to \mathbb{C}^{r+1}$ be the map from Example 3.1.4. The image of the interval $[0,1]$ in Δ_r is not a Zariski closed set. The Zariski closure of $\phi([0,1])$ equals $\phi(\mathbb{C})$. This is because $\phi(\mathbb{C})$ is Zariski closed, and the Zariski closure of $[0,1] \subseteq \mathbb{C}$ is all of $\mathbb{C}$.

Note that the Zariski closure of $[0,1]$ is not $\mathbb{R} \subseteq \mathbb{C}$. While $\{z \in \mathbb{C} : z = \overline{z}\}$ is not a variety when considered as a subset of $\mathbb{C}$, since complex conjugation is not a polynomial operation.

If we only care about subsets of Δ_r, then $\phi([0,1])$ is equal to the Zariski closure intersected with the probability simplex. In this case, the variety gives a good approximation to the statistical model of binomial random variables.

The complement of a Zariski closed set is called a Zariski open set. The Zariski topology is the topology whose closed sets are the Zariski closed sets. !! The Zariski topology is a much coarser topology than the standard topology, as the following example illustrates.

Example 3.2.10 (Mixture of binomial random variables)**.** Let $\mathcal{P} \subseteq \Delta_{r-1}$ be a family of probability distributions (i.e., a statistical model) and let $s > 0$ be an integer. The sth *mixture model* $\mathrm{Mixt}^s(\mathcal{P})$ consists of all probability distributions which arise as convex combinations of s distributions in $\mathcal{P}$.

That is,

$$\mathrm{Mixt}^s(\mathcal{P}) \quad = \quad \left\{ \sum_{j=1}^{s} \pi_j p^{(j)} : \pi \in \Delta_{s-1} \text{ and } p^{(j)} \in \mathcal{P} \text{ for all } j \right\}.$$

Mixture models allow us to build a complex model from a simple model $\mathcal{P}$. The motivation for modeling with a mixture model will be explained in Chapter 14.

The special case of mixtures of binomial random variables arises in Bayesian statistics through its connection to exchangeability via De Finetti's theorem (see, e.g., [**DF80**]). Focusing on the case $r = 2$ and $s = 2$, we see that $\mathrm{Mixt}^2(\mathcal{P})$ is precisely the region below the curve in Figure 3.1.1. Since $\mathrm{Mixt}^2(\mathcal{P})$ is a two-dimensional subset of Δ_2, we see that the Zariski closure of $\mathrm{Mixt}^2(\mathcal{P})$ is the entire plane $V(p_0+p_1+p_2-1)$. Thus, the intersection of the Zariski closure and the probability simplex Δ_2 is not the mixture model. This example illustrates the fact that in some situations in algebraic statistics, it might not be sufficient to only consider varieties. Indeed, sometimes constraints using inequalities are necessary. In this example, the mixture model is the set

$$\Delta_2 \cap \left\{ p : \det \begin{pmatrix} 2p_0 & p_1 \\ p_1 & 2p_2 \end{pmatrix} \geq 0 \right\}.$$

We will return to the issues of real algebraic geometry in later chapters.

Building on the Nullstellensatz and the notion of Zariski closures, there is a natural correspondence between radical ideals on the one hand and varieties on the other.

Theorem 3.2.11 (Ideal-variety correspondence)**.** *Let $\mathbb{K}$ be an algebraically closed field. Then the maps*

$$V : \left\{ I \subseteq \mathbb{K}[p] \;:\; I = \sqrt{I} \right\} \to \{ W \subseteq \mathbb{K}^r \;:\; W = V(J) \text{ for some } J \subseteq \mathbb{K}[p] \},$$

$$I : \{ W \subseteq \mathbb{K}^r \;:\; W = V(J) \text{ for some } J \subseteq \mathbb{K}[p] \} \to \left\{ I \subseteq \mathbb{K}[p] \;:\; I = \sqrt{I} \right\}$$

are inclusion-reversing bijections between the set of radical ideals and the set of varieties.

Note that the condition that $\mathbb{K}$ is algebraically closed is necessary (see Exercise 3.4).

Proof. The fact that $V(I(V)) = V$ for V a variety follows from the definition of variety. On the other hand, the Nullstellensatz shows that $I(V(I)) = I$ if I is a radical ideal and $\mathbb{K}$ is algebraically closed. This shows that V and I are bijections and inverses of each other. To see that these maps are inclusion reversing, note that if $I \subseteq J$, then $V(J) \subseteq V(I)$, since in the definition of variety $V(J)$ there are more constraints than in $V(I)$, and hence fewer points can satisfy these conditions. $\square$

Theorem 3.2.11 is the Rosetta stone for translating between algebra and geometry. Natural operations on the ideal or variety side lead to natural related operations on the opposing side. For example consider the sum of two ideals:

$$I + J := \{f + g \ : \ f \in I \text{ and } g \in J\}.$$

If $I = \langle \mathcal{F} \rangle$ and $J = \langle \mathcal{G} \rangle$, then $I + J = \langle \mathcal{F} \cup \mathcal{G} \rangle$. This observation immediately leads to a geometric consequence on the level of varieties.

Proposition 3.2.12. *Let I and J be ideals of $\mathbb{K}[p]$. Then*

$$V(I + J) = V(I) \cap V(J).$$

If $\mathbb{K}$ is an algebraically closed field and V and W are arbitrary subsets of $\mathbb{K}^r$, then

$$I(V \cap W) = \sqrt{I(V) + I(W)}.$$

Proof. It suffices to show the first equation, since the second follows by the ideal-variety correspondence. Suppose that $\mathbf{a} \in V(I + J)$. Then $f(\mathbf{a}) = 0$ for all $f \in I$ and for all $f \in J$. This implies that $\mathbf{a} \in V(I) \cap V(J)$ and implies the containment $V(I + J) \subseteq V(I) \cap V(J)$. On the other hand, if $\mathbf{a} \in V(I) \cap V(J)$, it satisfies all polynomials in I and in J, and hence all polynomials in $I + J$. $\square$

An analogous result holds connecting intersections of ideals and unions of varieties. The proof is left as an exercise.

Proposition 3.2.13. *Let I and J be ideals of $\mathbb{K}[p]$. Then*

$$V(I \cap J) = V(I) \cup V(J).$$

If V and W are arbitrary subsets of $\mathbb{K}^r$, then

$$I(V \cup W) = I(V) \cap I(W).$$

3.3. Gröbner Bases

Gröbner bases are the algebraic tool for studying ideals computationally. The most basic use for Gröbner bases is to answer the *ideal membership problem*; that is, to decide whether or not a given polynomial f belongs to a particular ideal I. The fact that Gröbner bases solve the ideal membership problem allows for a great many further applications. We highlight some of these uses in this chapter and defer other applications until later in the book.